

Ali Nikkhah

Tehran, Iran | +98 991 296 3951 | alinkkh9@gmail.com | alinikkhah2001.github.io | linkedin.com/in/alinikkhah2001

TECHNICAL SKILLS

Languages: Python (Asyncio, Pydantic), Go, C++, SQL, Bash, TypeScript, Node.js
Deep Learning & Core AI: PyTorch, JAX, TensorFlow, Hugging Face, Time-Series, ONNX Runtime
Agentic AI & Advanced RAG: LangGraph, LangChain, LlamaIndex, CrewAI, AutoGen, HyDE, Graph RAG, KAG
Vision & Audio Systems: OpenCV, BLIP, Vision Transformers (ViT), T5, Whisper, wav2vec2.0, Librosa, ESPnet
Data Engineering & Lakes: Apache Spark, Trino, ClickHouse, Airflow, PeerDB, Apache Kafka, RabbitMQ, Parquet
MLOps & Infrastructure: Ray, Kubeflow, Triton Inference Server, TensorFlow Serving, TensorRT-LLM, TF Lite
DevOps & Automation: Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, GitLab CI
Monitoring, Governance & BI: Prometheus, Grafana, ELK Stack, Metabase, Weights & Biases, MLflow, DVC

PROFESSIONAL EXPERIENCE

Digikala

Tehran, Iran

Senior AI Engineer

Nov 2024 - Aug 2025

- Led the engineering and deployment of the enterprise central intelligence layer, transitioning production operations from baseline semantic matching to stateful, multi-agent automated systems.
- Architected Modular Control Pipelines (MCP) to decouple text ingestion, chunking, retrieval, and synthesis, enabling independent hot-swapping of underlying LLMs and vector database indexes without service interruption.
- Designed and integrated hybrid retrieval engines executing concurrent dense semantic vector searches and sparse lexical keyword lookups, fusing processing streams via Reciprocal Rank Fusion (RRF) to preserve exact keyword and serial identifier matches.
- Engineered autonomous, stateful agentic routing networks using cyclic directed graphs to manage multi-turn session persistence, dynamic context evaluation, and automated query rewriting loops.
- Built asynchronous backend serving APIs leveraging non-blocking event loops and multi-tier hierarchical caching layers to isolate model execution overhead from raw client request streams.

Turquoise Digital

Tehran, Iran

Senior Data Engineer / Data Scientist

Aug 2025 - Mar 2026

- Engineered an enterprise financial agentic chatbot utilizing state graphs and dynamic tool function-calling to automate multi-turn client support, financial literacy instruction, and interactive portfolio querying.
- Implemented secure data-access verification layers using data validation boundaries to safely execute real-time queries against user equity balances and internal algorithmic calculation suites.
- Developed a low-latency telephony audio intelligence pipeline by integrating streaming transcription engines (Whisper, wav2vec2.0) with an asynchronous processing network to capture, transcribe, and route live customer service requests.

SELECTED PROJECTS & RESEARCH

Heuristic-Driven Forex Market Prediction Engine

Personal Research & Development

Lead Quantitative Developer

2024

- Developed a predictive quantitative modeling pipeline for foreign exchange markets that mathematically maps professional trader perspectives, translating chart components like dynamic trendlines and support/resistance zones into distinct feature arrays.
- Engineered a sequence-to-sequence deep learning configuration to align technical chart indicators with quantitative market microstructure parameters to identify potential trend reversals and automate trading execution loops.

Dynamic AI Portfolio Allocation & Recommendation Framework

Turquoise Digital

Core Architect

2025

- Built a highly scalable asset recommendation system that parsed risk-assessment questionnaires via an interactive user loop to compute multi-variable risk tolerances and execute automated batch trading cycles.
- Aligned historical time-series pricing data with streaming market sentiment news embeddings, utilizing deep temporal sequence networks and time-series transformer architectures to project future valuation curves alongside explicit statistical confidence intervals.

EDUCATION

Sharif University of Technology

Tehran, Iran

B.Sc. Electrical Engineering - Digital Systems (GPA: 3.65, Major: 3.91)

Sept 2020 - Jul 2024